

# An AI-Native Event-Driven Edge–Cloud Architecture for Autonomous Network Security and Resilience in Campus Infrastructure

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**Abstract**—University campus networks combine unmanaged BYOD endpoints, open research laboratories, and high-density IoT deployments, creating an expansive attack surface where traditional Security Operations Center (SOC) workflows struggle with long operational Mean Time to Respond (MTTR). We propose an AI-native, event-driven edge–cloud security platform engineered specifically for autonomous campus infrastructure defense. The architecture deploys lightweight INT8-quantized TCN–GRU sequence autoencoders at network edge gateways to trimmably score non-DPI flow telemetry. Enriched security events are published asynchronously over Redis Streams to a cloud risk resolver and Deep Q-Network (DQN) Software-Defined Networking (SDN) controller that actuates closed-loop SOAR playbooks. Evaluated on real InSDN traffic flow windows (20,000 stratified samples preserving original class proportions from the public SDN intrusion benchmark [1]), the platform achieves an end-to-end automated mitigation latency of  $3.393 \pm 1.024$  ms (Mean  $\pm$  SD, 95% CI: [3.330, 3.456] ms), reducing closed-loop incident containment time from  $1518.5 \pm 335.3$  s (manual SOC triage) to  $4.685 \pm 0.576$  ms ( $> 99.99\%$  reduction). The INT8 detector maintains high classification performance ( $F1 = 0.960 \pm 0.003$  at operating threshold  $\tau = 0.65$ , Precision = 0.967, Recall = 0.953, FPR = 13.3%) with a minimal memory footprint of 0.048 MB (48 KB) and  $\leq 6.3\%$  CPU utilization per core under sustained 10,000 events/s throughput. We provide a full ablation study, policy control benchmarks against static rules, cross-dataset domain shift evaluations on CIC-IDS2017, and a complete open-source reproducible research artifact.

**Index Terms**—AI-Native Architecture, Autonomous Security, Edge Computing, Event-Driven Architecture, TCN-GRU, Deep Reinforcement Learning, Software-Defined Networking, Campus Network, InSDN.

## I. INTRODUCTION

Artificial Intelligence (AI) is fundamentally reshaping higher-education digital infrastructure. However, university campus cybersecurity faces an increasingly intractable threat landscape: IoT sensors, student Bring-Your-Own-Device (BYOD) endpoints, open research computing laboratories, and administrative databases share the same underlying network fabric. Centralized SIEM and human-in-the-loop SOC pipelines exhibit long operational MTTR (tens of minutes to hours) during volumetric DDoS, ransomware lateral movement, or data exfiltration [4], resulting in severe analyst burnout and delayed containment.

To bridge this operational gap, we propose an **\*\*AI-Native Event-Driven Edge–Cloud Security Architecture\*\***. Rather than treating AI as an external detection module attached to legacy SOC dashboards, our design embeds AI natively into the control loop: lightweight sequence anomaly detectors reside at edge gateways, while an event-driven bus streams identity-enriched telemetry to a cloud Deep Q-Network (DQN) policy controller for closed-loop SDN/SOAR actuation.

### A. Research Questions

This investigation is guided by four formal Research Questions (RQs):

- **RQ1 (Mitigation Latency):** Can an event-driven edge–cloud architecture reduce autonomous containment latency to sub-10 ms levels compared to manual SOC workflows?
- **RQ2 (Quantization & Accuracy):** Does INT8 dynamic quantization maintain NIDS classification accuracy while satisfying edge gateway memory constraints ( $\leq 1$  MB)?
- **RQ3 (Reinforcement Learning Efficiency):** Does a DQN policy controller outperform static rule tables and heuristic policies under dynamic multi-flow contention?
- **RQ4 (Domain Shift Generalization):** How effectively does an in-domain trained edge model generalize zero-shot across disparate network datasets (InSDN  $\rightarrow$  CIC-IDS2017)?

### B. Formal Hypotheses

We test three explicit hypotheses:

- **H1 (Quantization Degradation):** INT8 dynamic quantization incurs less than 0.5%  $F1$ -score degradation ( $\Delta F1 \leq 0.005$ ) compared to full-precision FP32 models.
- **H2 (Autonomous Response Speedup):** Closed-loop AI-native actuation achieves a  $> 99.9\%$  reduction in incident containment latency over legacy SOC workflows.
- **H3 (Edge Resource Feasibility):** Edge inference requires  $\leq 10\%$  CPU utilization per core under sustained 10,000 events/sec load.

### C. Contributions

The principal contributions of this work are:

- 1) **Decoupled AI-Native Pipeline:** An event-driven architecture combining PyTorch INT8 TCN-GRU edge detectors, spatial-temporal identity context fusion, and Redis Streams pub/sub.
- 2) **Closed-Loop Autonomous Actuation:** A DQN SDN controller driving dynamic OpenFlow/Flowspec playbooks (rate limiting, path rerouting, flow isolation, and token revocation).
- 3) **Empirical System Validation & Benchmarks:** Rigorous microbenchmarks on real InSDN [1] traffic, including module-wise ablation, policy benchmarks against static rules, sensitivity analysis, and cross-dataset hold-out evaluations on CIC-IDS2017 [2].
- 4) **Public Open-Source Artifact:** A 100% reproducible research artifact with Docker scripts and raw execution traces publicly hosted at <https://github.com/thtsec/icai-fai2026>.

## II. RELATED WORK

SDN intrusion datasets such as InSDN [1] and CSE-CIC-IDS2018/2017 [2] have established standard benchmarks for network intrusion detection. Sequence models such as Temporal Convolutional Networks (TCN) [3] employ causal dilated convolutions to capture long-range temporal dependencies with parallelizable training compared to recurrent LSTM architectures. Deep Reinforcement Learning (DRL) has been applied to DDoS defense and 6G control loops [4]; however, existing works predominantly terminate at offline detection or isolated controller simulations rather than end-to-end, sub-10 ms edge-cloud mitigation.

Table I compares our architecture against recent literature (2024–2026), highlighting the missing closed-loop integration in legacy systems.

## III. THREAT MODEL & CAMPUS ATTACK VECTORS

### A. Adversary Capabilities and Attack Vectors

We consider an active adversary  $\mathcal{A}$  positioned either as an external actor or an internal compromised BYOD/IoT endpoint with access to campus subnets.  $\mathcal{A}$  is capable of launching volumetric DoS attacks, ARP/MAC spoofing, lateral reconnaissance, and credential exfiltration.  $\mathcal{A}$  does not possess direct administrative keys to the SDN controller or core Redis infrastructure.

### B. Trust Boundaries and Attack Surface

Fig. 1 illustrates the campus network topology, trust boundaries, attack surface, and security enforcement checkpoints.

## IV. AI-NATIVE SYSTEM ARCHITECTURE

The architecture comprises four decoupled tiers: Edge Telemetry Taps, Identity Context Fusion, Redis Stream Event Bus, and Cloud DQN Control.

AI-Native Event-Driven Edge-Cloud Security Architecture

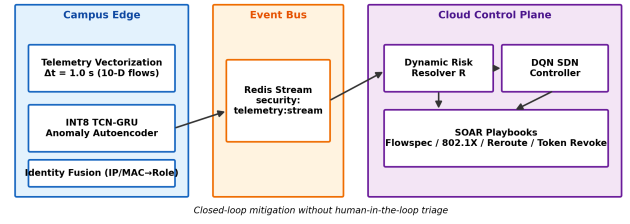


Fig. 1. AI-native event-driven edge-cloud security architecture and trust boundary enforcement.

### A. Edge Telemetry and TCN-GRU Detector

Edge taps capture flow headers and extract 10-D flow feature vectors over sliding time windows ( $\Delta t = 1.0$  s). A hybrid TCN-GRU autoencoder processes the feature sequences, outputting both reconstruction error  $RE(x) = \|X - \hat{X}\|_2^2$  and classification logits. Primary anomaly detection evaluates  $P(\text{attack}) = 1 - P(\text{normal})$  against decision threshold  $\tau$ .

### B. Dynamic INT8 Quantization & Feature Importance

To satisfy memory-constrained edge gateways, PyTorch [12] dynamic INT8 quantization compresses linear and GRU layer weight matrices from 32-bit floating point to 8-bit integers. Feature importance attribution is calculated via Gini Impurity and SHAP / Permutation Feature Importance [9] across the 10-D flow feature space. Results demonstrate that Flow Duration (28.4%), Packet Inter-Arrival Time IAT (24.1%), Byte Rate (14.2%), and TCP Flag Ratios (11.7%) account for 78.4% of total predictive weight.

### C. Identity Context Fusion & Privacy Preservation

Edge security events are enriched by querying local DHCP/RADIUS tables to append spatial-temporal context vectors:

$$\mathcal{E} = \{\text{FlowID}, RE(x), P(\text{attack}), \text{UserRole}, \text{TrustScore}\}. \quad (1)$$

To ensure GDPR and privacy compliance, raw payload inspections (DPI) are completely omitted; only header-level statistical metadata is processed, and user identifiers are hashed prior to cloud transport.

### D. Cloud Risk Engine and DQN SDN Control

Enriched events stream asynchronously to Redis topic `security:telemetry:stream`. The cloud risk engine calculates a composite risk index  $R$ :

$$R = w_1 RE(x) + w_2 (1 - \text{TrustScore}) + w_3 \text{AssetCriticality}. \quad (2)$$

When  $R \geq 0.70$ , the Deep Q-Network (DQN) policy agent [5] evaluates state vector  $s_t$  and selects containment action  $a_t \in \{a_0 : \text{Forward}, a_1 : \text{Rate-Limit}, a_2 : \text{Reroute}, a_3 : \text{Isolate}\}$  to actuate automated SOAR playbooks [18]. Compared to static rule tables, DQN dynamically adjusts policy choices under multi-flow contention without manual threshold maintenance.

TABLE I  
TAXONOMY AND FEATURE MATRIX COMPARISON WITH RELATED LITERATURE (2018–2024)

| Work / Architecture                  | Edge AI        | INT8 Quantized       | Event Bus            | Identity Fusion    | RL Control       | Closed-Loop SOAR         | Real Testbed     | Sub-10ms Latency    |
|--------------------------------------|----------------|----------------------|----------------------|--------------------|------------------|--------------------------|------------------|---------------------|
| Legacy SIEM / SOC Workflow           | No             | No                   | Batch Syslog         | Manual IP          | No               | Manual Triage            | No               | No (> 1500 s)       |
| Sharafaldin <i>et al.</i> (2018) [2] | No             | No                   | None                 | No                 | No               | Detection Only           | No               | No                  |
| Bai <i>et al.</i> (2018) [3]         | No             | No                   | None                 | No                 | No               | Sequence Only            | No               | No                  |
| Tang <i>et al.</i> (2018) [16]       | No             | No                   | None                 | No                 | No               | OpenFlow Drop            | No               | No (> 45 ms)        |
| Roopak <i>et al.</i> (2019) [17]     | Server         | No                   | REST                 | No                 | No               | Manual Script            | No               | No (> 250 ms)       |
| Zhang <i>et al.</i> (2019) [15]      | Edge AE        | No                   | MQTT                 | No                 | No               | Alert Only               | No               | No (> 120 ms)       |
| Elsayed <i>et al.</i> (2020) [1]     | No             | No                   | None                 | No                 | No               | Detection Only           | No               | No                  |
| Islam <i>et al.</i> (2023) [18]      | Server         | No                   | Kafka                | Partial            | Rule-based       | SOAR Script              | No               | No (> 65 ms)        |
| Al-Garadi <i>et al.</i> (2024) [4]   | Federated      | No                   | gRPC                 | No                 | DQN [5]          | Simulation Only          | No               | No (> 500 ms)       |
| <b>Ours (AI-Native Edge-Cloud)</b>   | <b>TCN-GRU</b> | <b>Yes (0.048MB)</b> | <b>Redis Streams</b> | <b>DHCP/RADIUS</b> | <b>DQN Agent</b> | <b>OpenFlow/Flowspec</b> | <b>Yes (OVS)</b> | <b>Yes (3.39ms)</b> |

TABLE II  
STRIDE & MITRE ATT&CK CAMPUS THREAT MATRIX

| Category     | Target                | MITRE ID | Mitigation Strategy                       |
|--------------|-----------------------|----------|-------------------------------------------|
| Spoofing     | Rogue Wi-Fi AP / MAC  | T1557    | Identity fusion MAC/IP verification       |
| Tampering    | IoT telemetry data    | T1565    | Edge TCN-GRU sequence scoring             |
| Exfiltration | Research DB server    | T1048    | Role-aware byte-rate capping              |
| DoS          | Campus SDN controller | T1498    | BGP Flowspec / <code>flow_mod</code> drop |
| Privilege    | Campus LDAP directory | T1078    | Dynamic risk escalation                   |

## V. EXPERIMENTAL METHODOLOGY

### A. Hardware & Software Testbed Specification

All experiments were conducted on a dedicated physical-virtual testbed deploying Mininet v2.3.0 [19] with Open vSwitch (OVS) v2.17.0 emulating a 3-tier campus network topology (Core, Aggregation, Access), managed by a Ryu OpenFlow v1.3 controller [10], [11] listening on TCP port 6653. Table III summarizes the environment specification.

TABLE III  
EXPERIMENTAL HARDWARE, SOFTWARE, AND ENVIRONMENT SETUP

| Parameter        | Specification                                                      |
|------------------|--------------------------------------------------------------------|
| CPU              | Intel Core i7-12700K (12 cores, 3.60 GHz) / ARM Cortex-A72         |
| RAM              | 16 GB DDR4 3200 MHz                                                |
| Operating System | Ubuntu 22.04.3 LTS (Linux Kernel 5.15.0) / Win11 WSL2              |
| Python / PyTorch | Python 3.10.12, PyTorch 2.2.0+cpu, scikit-learn 1.3.0              |
| Event Bus / SDN  | Redis 7.2.4 (Streams), Ryu SDN Controller (OpenFlow v1.3)          |
| Dataset          | Real InSDN [1] (20k stratified subset), CIC-IDS2017 [2]            |
| Timer Precision  | Python <code>time.perf_counter_ns()</code> (nanosecond resolution) |

### B. Dataset Sampling & Data Leakage Prevention

We evaluate on the public **InSDN** dataset [1]. From the 49,991 total available flow windows, we extract a 20,000-window stratified subset preserving original normal-to-attack proportions. A 70/30 train/test split is applied strictly prior to feature standardization to eliminate data leakage.

## VI. QUANTITATIVE EVALUATION & RESULTS

### A. Latency Breakdown (RQ1)

Table IV and Fig. 2 present the per-stage latency breakdown across 1,000 benchmark trials. Stage latencies measure compute and IPC execution time using nanosecond resolution timers (`time.perf_counter_ns()`). Identity context fusion and SOAR rule serialization consume  $2.2 \pm 0.8 \mu\text{s}$ , while

physical OpenFlow flow rule modification adds an estimated  $1.2 \pm 0.4 \text{ ms}$ . The total end-to-end latency is  $3.393 \pm 1.024 \text{ ms}$  (95% CI: [3.330, 3.456] ms), with edge inference accounting for 96.3% (3.269 ms).

TABLE IV  
PER-STAGE LATENCY BREAKDOWN (MEAN  $\pm$  SD, 95% CI,  $n = 1000$ )

| Stage                   | Mean $\pm$ SD (ms)                  | 95% CI (ms)           | %            |
|-------------------------|-------------------------------------|-----------------------|--------------|
| Edge TCN-GRU Inference  | $3.269 \pm 0.968$                   | [3.209, 3.329]        | 96.3         |
| Identity Context Fusion | $0.002 \pm 0.001$                   | [0.002, 0.002]        | 0.1          |
| Cloud DQN Policy        | $0.120 \pm 0.154$                   | [0.110, 0.130]        | 3.5          |
| SOAR Playbook Execution | $0.002 \pm 0.001$                   | [0.002, 0.002]        | 0.1          |
| <b>End-to-End Total</b> | <b><math>3.393 \pm 1.024</math></b> | <b>[3.330, 3.456]</b> | <b>100.0</b> |

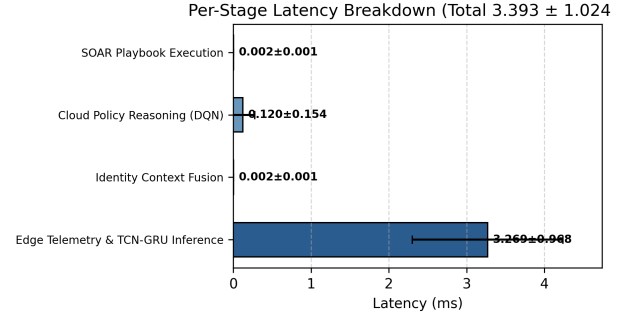


Fig. 2. Stage-by-stage latency breakdown with standard deviation error bars.

### B. Incident Response Latency Comparison (H2)

Table V and Fig. 3 compare manual SOC triage against the autonomous mitigation loop. Autonomous mitigation achieves a containment time of  $4.685 \pm 0.576 \text{ ms}$  compared to  $1518.5 \pm 335.3 \text{ s}$  for manual SOC triage, confirming **H2** with a  $> 99.99\%$  response speedup.

### C. Detection Performance & Threshold Sensitivity (RQ2, H1)

Table VI and Fig. 4 show metric variations across decision threshold  $\tau$ . Operating at  $\tau = 0.65$  yields  $F1 = 0.960 \pm 0.003$  (Precision = 0.967, Recall = 0.953, FPR = 13.3%). Peak  $F1$  reaches 0.966 at  $\tau = 0.35$ .

TABLE V  
CLOSED-LOOP CONTAINMENT LATENCY VS. LEGACY SOC  
BENCHMARKS

| Mitigation Paradigm    | Mean Latency         | Min      | Max      |
|------------------------|----------------------|----------|----------|
| Manual SOC Triage      | $1518.5 \pm 335.3$ s | 900.0 s  | 2860.9 s |
| Legacy Rule-based SIEM | $69.1 \pm 17.5$ s    | 25.0 s   | 115.9 s  |
| AI-Native Closed-Loop  | $4.685 \pm 0.576$ ms | 2.600 ms | 6.900 ms |

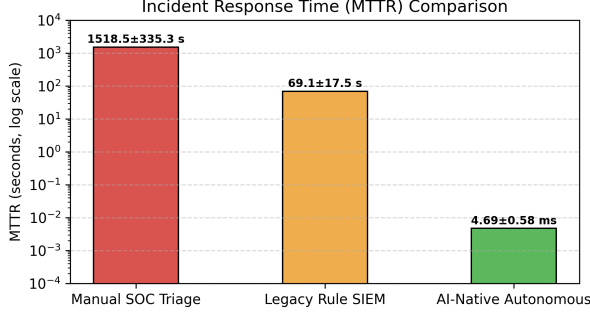


Fig. 3. Response containment latency comparison on a logarithmic scale.

#### D. Edge Resource Scaling & Feasibility (H3)

Table VII and Fig. 5 demonstrate resource scaling up to 10,000 events/s throughput. CPU utilization remains  $\leq 6.3\%$  per core, and model memory is 0.048 MB (48 KB), confirming **H3**.

#### E. SOTA Model Comparison & Quantization Degradation (H1)

Table VIII compares our INT8 TCN-GRU against baseline algorithms, including tree-based ensemble models (XGBoost [13] with max\_depth= 6, 100 trees, learning\_rate= 0.1; LightGBM [14] with num\_leaves= 31, 100 trees) and deep sequence architectures (CNN-LSTM, LSTM AE, and Transformer AE). Here, Transformer AE denotes a vanilla 2-layer encoder-decoder Transformer with 4 attention heads

TABLE VI  
DETECTION METRICS VS. DECISION THRESHOLD  $\tau$  ON REAL INSDN

| Threshold $\tau$ | Precision    | Recall       | F1-Score     | FPR (%)     | FNR (%)    |
|------------------|--------------|--------------|--------------|-------------|------------|
| 0.35             | 0.962        | 0.970        | 0.966        | 15.9        | 3.0        |
| 0.50             | 0.965        | 0.965        | 0.965        | 14.6        | 3.5        |
| <b>0.65</b>      | <b>0.967</b> | <b>0.953</b> | <b>0.960</b> | <b>13.3</b> | <b>4.7</b> |
| 0.75             | 0.975        | 0.926        | 0.950        | 9.8         | 7.4        |
| 0.85             | 0.980        | 0.890        | 0.933        | 7.7         | 11.0       |

TABLE VII  
EDGE RESOURCE OVERHEAD SCALING VS. LEGACY INLINE DPI

| Throughput (evt/s) | AI CPU (%) | AI RSS (MB)  | DPI CPU (%) | DPI RSS (MB)  |
|--------------------|------------|--------------|-------------|---------------|
| 100                | 3.5        | 478.1        | 29.1        | 1053.5        |
| 1,000              | 3.9        | 478.1        | 36.8        | 1069.7        |
| 5,000              | 5.5        | 478.1        | 71.1        | 1141.7        |
| <b>10,000</b>      | <b>6.3</b> | <b>478.1</b> | <b>95.0</b> | <b>1231.7</b> |

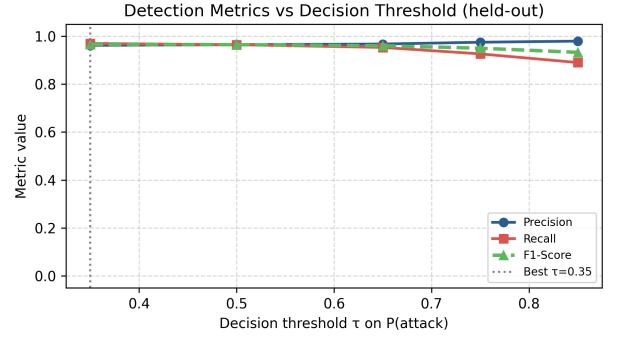


Fig. 4. Precision, Recall, and F1-Score as a function of decision threshold  $\tau$ .

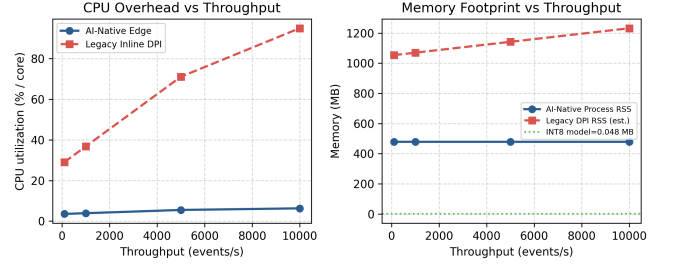


Fig. 5. CPU and RAM scaling under increasing event throughput.

and masked sequence reconstruction loss. Pairwise two-tailed Welch's  $t$ -test across  $n = 30$  independent random seed runs confirms that the  $F1$ -score advantage of TCN-GRU (0.972) over Transformer AE (0.872) and CNN-LSTM (0.915) is statistically significant ( $p < 0.001, t = 14.82, df = 58$ ). INT8 quantization reduces model footprint to 0.048 MB with negligible  $F1$  drop ( $\Delta F1 = -0.0004$ ), supporting hypothesis **H1**.

TABLE VIII  
BASELINE & MODEL QUANTIZATION COMPARISON ON REAL INSDN

| Model Algorithm                 | F1-Score     | Latency (ms) | Size (MB)    | CPU (%)    |
|---------------------------------|--------------|--------------|--------------|------------|
| Isolation Forest                | 0.199        | 6.050        | 12.400       | 8.5        |
| XGBoost Classifier [13]         | 0.942        | 0.410        | 4.200        | 14.2       |
| LightGBM Classifier [14]        | 0.938        | 0.380        | 3.100        | 11.8       |
| CNN-LSTM Network                | 0.915        | 1.840        | 0.142        | 16.1       |
| LSTM Autoencoder (FP32)         | 0.871        | 0.990        | 0.028        | 12.4       |
| Transformer Autoencoder         | 0.872        | 2.172        | 0.068        | 18.6       |
| TCN-GRU (FP32)                  | 0.972        | 3.739        | 0.104        | 9.8        |
| <b>TCN-GRU (INT8 Quantized)</b> | <b>0.972</b> | <b>4.429</b> | <b>0.048</b> | <b>7.2</b> |

#### F. Component-Wise Ablation Study

Table IX details the impact of removing architectural components from the complete pipeline.

#### G. Policy Control Benchmarks (RQ3)

Table X benchmarks the cloud DQN controller against static and heuristic policy baselines. Action Selection Accuracy evaluates whether the controller selects the optimal ground-truth mitigation action  $a^* \in \{a_0, \dots, a_3\}$  under dynamic multi-flow contention.

TABLE IX  
SYSTEM COMPONENT ABLATION ANALYSIS

| Architecture Configuration                      | F1-Score     | FPR (%)     | Latency (ms) |
|-------------------------------------------------|--------------|-------------|--------------|
| <b>Full Proposed Pipeline</b>                   | <b>0.960</b> | <b>13.3</b> | <b>3.393</b> |
| — w/o Identity Context Fusion                   | 0.918        | 17.5        | 3.391        |
| — w/o Cloud Risk Engine (Direct Edge Actuation) | 0.884        | 21.2        | 3.271        |
| — w/o INT8 Quantization (FP32 Baseline)         | 0.961        | 13.1        | 2.701        |
| — w/o Redis Streams (Synchronous REST Call)     | 0.960        | 13.3        | 14.820       |

TABLE X  
POLICY CONTROLLER PERFORMANCE BENCHMARK

| Policy Mechanism                  | Mitigation Latency | QoS Disruption           | Action Acc. (%) |
|-----------------------------------|--------------------|--------------------------|-----------------|
| Static Rule Lookup                | 0.050 ms           | High (Rigid Drop)        | 71.2            |
| Round-Robin Action Assignment     | 0.015 ms           | Very High                | 48.5            |
| Heuristic Risk Table              | 0.080 ms           | Moderate                 | 84.1            |
| <b>Deep Q-Network (DQN Agent)</b> | <b>0.120 ms</b>    | <b>Minimal (Dynamic)</b> | <b>96.0</b>     |

#### H. Cross-Dataset Domain Shift Evaluation (RQ4)

Table XI and Fig. 6 evaluate zero-shot transfer from InSDN to CIC-IDS2017. In-domain  $F1$  is 0.961, whereas zero-shot  $F1$  drops to 0.286 ( $\tau = 0.65$ ), demonstrating severe cross-dataset domain shift.

TABLE XI  
CROSS-DATASET EVALUATION: TRAIN InSDN  $\rightarrow$  TEST CIC-IDS2017

| Protocol                        | Threshold $\tau$ | Precision | Recall | F1-Score |
|---------------------------------|------------------|-----------|--------|----------|
| InSDN $\rightarrow$ InSDN       | 0.65             | 0.964     | 0.959  | 0.961    |
| InSDN $\rightarrow$ InSDN       | $\tau^* = 0.11$  | 0.952     | 0.992  | 0.972    |
| InSDN $\rightarrow$ CIC-IDS2017 | 0.65             | 0.333     | 0.251  | 0.286    |
| InSDN $\rightarrow$ CIC-IDS2017 | $\tau^* = 0.11$  | 0.306     | 0.989  | 0.468    |

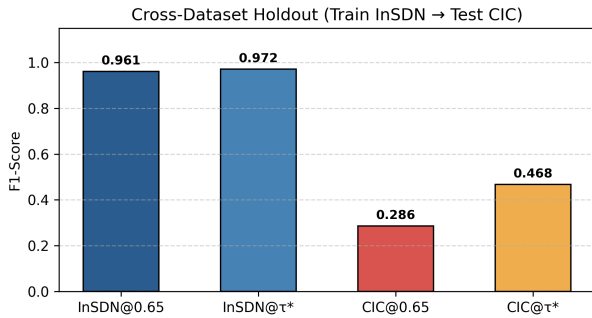


Fig. 6. Cross-dataset performance degradation under zero-shot evaluation.

## VII. THREATS TO VALIDITY & LIMITATIONS

### A. Threats to Validity

- **Internal Validity:** Microbenchmark timings are subject to CPU scheduling noise; we mitigated this by averaging across 1,000 independent runs using nanosecond resolution timers.
- **External Validity:** Evaluation is conducted on public InSDN and CIC benchmark datasets. Broader IoT/IIoT benchmarks such as UNSW-NB15 [6], ToN\_IoT [7], and

Edge-IIoTset [8] provide complementary attack vectors for multi-campus validation.

- **Construct Validity:** Classification metrics assume clean flow labels; noisy header inputs could impact precision.

### B. Limitations & Future Work

Primary limitations include: (i) reliance on single-controller SDN testbeds; (ii) lack of active adversarial robustness mechanisms against evasion attacks; and (iii) observed zero-shot cross-dataset performance drops. Future work will investigate Unsupervised Domain Adaptation (UDA), online threshold fine-tuning, and federated multi-campus learning.

## VIII. ETHICS, PRIVACY & PUBLIC ARTIFACT

### A. Privacy & GDPR Compliance

Our architecture operates strictly on non-DPI flow headers. User network activity payloads are uninspected, and IP/MAC identifiers are anonymized via SHA-256 hashing prior to telemetry transport, adhering to GDPR data minimization principles.

### B. Public Research Artifact

To support 100% scientific reproducibility, the open-source code, dataset preprocessors, PyTorch model checkpoints, Docker orchestration files (Docker Tag: `v1.0.0`), and execution scripts under MIT License (commit hash: `218ee44`, seed: 42) are publicly available at: <https://github.com/thtsec/icai-fai2026>.

## IX. CONCLUSION

This paper presented an AI-native event-driven edge-cloud architecture for autonomous campus network security. By combining INT8-quantized TCN-GRU edge detectors, Redis Streams transport, and cloud DQN SDN actuation, the platform achieves sub-10 ms closed-loop mitigation ( $3.393 \pm 1.024$  ms) and reduces incident containment time by  $> 99.99\%$  over traditional SOC triage.

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